

Kushal Agrawal

kushalag@andrew.cmu.edu | +1 (608) 895-1762 | [linkedin.com/in/kushalagrawal0](https://www.linkedin.com/in/kushalagrawal0) | kushalagrawal.net

EDUCATION

Carnegie Mellon University

Master of Science in Computer Engineering

Pittsburgh, PA

Aug 2026 – Dec 2027

University of Wisconsin–Madison

Bachelor of Science in Computer Engineering, GPA: 3.7

Madison, WI

Sep 2022 – May 2026

Coursework: Modern Computer Architecture, Digital System Design & Synthesis, Design-for-Test, CMOS Circuit Design, High-Performance Computing, Data Structures & Algorithms, Microprocessors, Deep Learning, Operating Systems

EXPERIENCE

Google

Madison, WI

Capstone Software Engineer Intern

Feb 2026 – May 2026

- Built IoTriage, a full-stack tool that scans a local network for IoT devices, identifies known vulnerabilities via the NIST NVD, and generates Gemini remediation guidance (Python, Nmap, Express, React).
- Built a rate-limit-aware NVD client with severity and false-positive filtering for accurate vulnerability rankings.

University of Wisconsin–Madison

Madison, WI

Undergraduate Teaching Assistant

Sep 2023 – May 2026

- Led office hours for 850+ students in Digital System Fundamentals, Machine Organization and Intro to Comp. Eng.
- Instructed students in ECE 352, ECE 354, and ECE 252 topics including digital logic design, processor architecture, RTL design, finite state machines, timing analysis, pipelining, C/assembly programming, and memory hierarchy.

Endress+Hauser Group

Greenwood, IN

Embedded Software Engineer Intern

May 2025 – Dec 2025

- Characterized Bluetooth 6.2 performance on the nRF54L15 via Channel Sounding and multi-device tests, analyzing ranging accuracy, packet reliability, and RF congestion effects for the SGC200 Bluetooth-to-cloud gateway.
- Engineered a Dockerized full-stack Netilion IIoT simulator (React/TypeScript, Python/FastAPI) with 30+ REST endpoints, 5 simulation patterns and 200+ health codes, streaming real-time data for hardware-free demos.

Software Engineer Intern

May 2024 – Aug 2024

- Designed a custom analog front-end and microcontroller interface to generate and read a 4–20 mA signal from an industrial temperature probe, enabling accurate current-to-temperature conversion for real-time IoT visualization.

PROJECTS

TradeMark – Dual-FPGA Trading Engine | *SystemVerilog, Tcl, Vivado, Python, PYNQ* | [GitHub](#)

- Built a closed-loop dual-FPGA trading system on two AMD AUP-ZU3 boards, one as a 16-symbol market/exchange and the other a pipelined trader, connected by a custom full-duplex PMOD link with hardware backpressure.
- Implemented a deterministic SystemVerilog trading loop in which a multi-regime market issues quotes, a pipelined trader with strategy and risk returns orders, and an exchange matches fills over a mesochronous framed link.
- Achieved ~80 ns tick-to-trade and ~2.07 μ s quote-to-fill, with latency and P&L tracked live via AXI-Lite registers.

Partial-Scan DFT Optimizer | *Tcl, Synopsys Design Compiler, TetraMAX, Verilog, STIL* | [GitHub](#)

- Architected a partial-scan DFT flow under ATE pin constraints, using SCOAP-based FF selection to cut scan overhead, and coverage-constrained ATPG to reach >85% stuck-at coverage with ~4 $\times$ fewer test patterns.
- Boosted a composite DFT metric by >3 $\times$ through optimizing test coverage, pattern count, and scan overhead.

5-Stage Pipelined RISC Processor | *Verilog, ModelSim, Synopsys Design Compiler* | [GitHub](#)

- Designed a 16-bit, 5-stage pipelined processor based on MIPS R2000, developing the data/control path with an optimized ALU, forwarding, hazard detection, branch prediction and a 2-way set-associative instruction/data cache.
- Built a cache-miss FSM and memory arbiter to refill 16-byte instruction/data cache blocks under contention.

FPGA Knight's Tour | *SystemVerilog, ModelSim, Synopsys Design Compiler, Quartus Prime* | [GitHub](#)

- Built an FPGA Knight's Tour robot with UART/Bluetooth and SPI inertial links, PID motor control, and IR tracking; achieved 333 MHz timing closure, 25% lower area (Synopsys DC), and 12% lower power via clock gating.
- Built self-checking SystemVerilog testbenches for functional/post-synthesis validation, achieving full code coverage.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Verilog, SystemVerilog, VHDL, Bash, Assembly, MATLAB, Tcl

Embedded: ARM/MIPS ISA, FreeRTOS, UART, SPI, I2C, BLE 6.2, MSP432, PSoC 6, nRF54L15, PYNQ

EDA & Verification: Vivado, Synopsys Design Compiler, TetraMAX, ModelSim, Quartus Prime, OpenROAD, Altium

Systems & Tools: Linux, Git, Docker, CMake, Make, GDB, CUDA, OpenMP, Slurm, FUSE, xv6